EE 66 Signals, Dynamics, and Information
Fall 2026 Discussion 0B

1. Complex Algebra (Review)

Let $a = 1 - i\sqrt{3}$ and $b = \sqrt{3} + i$.

(a) Show the numbers a and b in the complex plane below, marking the distance from the origin and angle with the real axis.

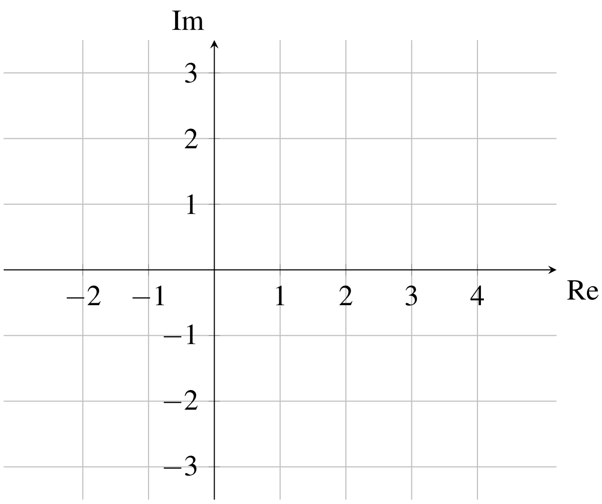

Figure 1: Complex Plane for a

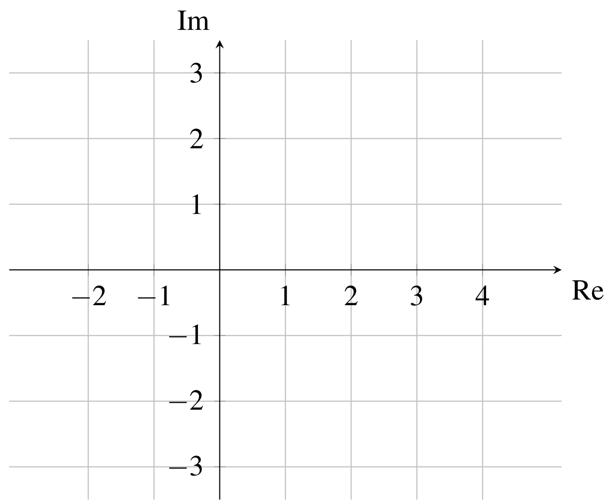

Figure 2: Complex Plane for b

Answer:

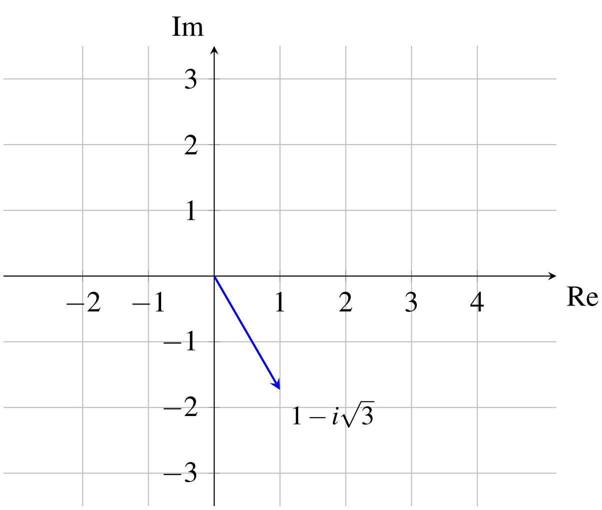

Figure 3: Complex Plane for $a = 1 - i\sqrt{3}$

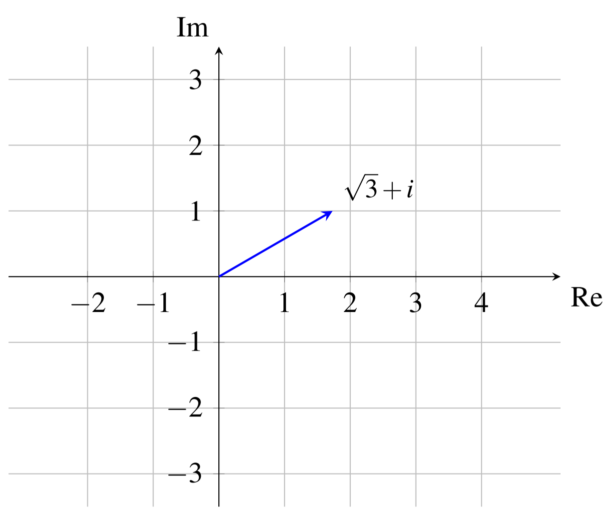

Figure 4: Complex Plane for b

(b) Multiply a with i, and plot the resulting expression a' below. Show that a' is equivalent to a rotated *counterclockwise* by $\frac{\pi}{2}$ or $90°$ in the complex plane. Multiply b with $-i$, and plot the resulting expression b' below. Show that b' is equivalent to b rotated *clockwise* by $\frac{\pi}{2}$ or $90°$ in the complex plane.

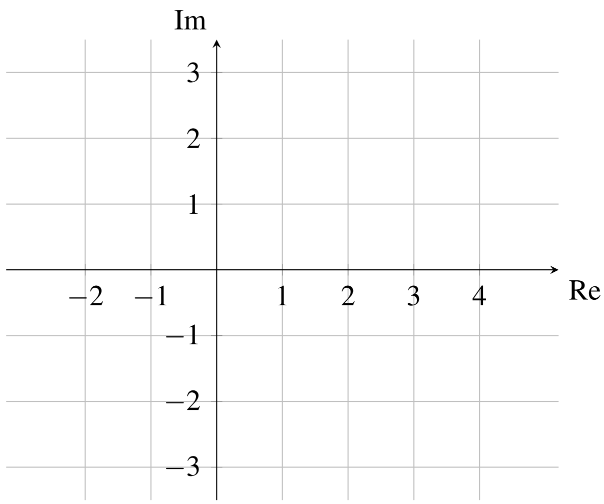

Figure 5: Complex Plane for a'

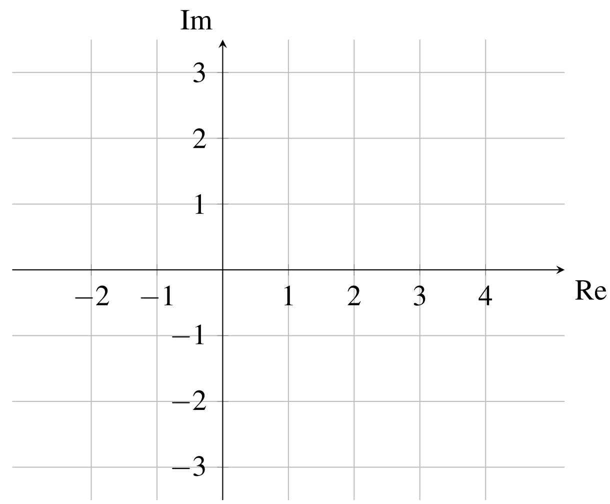

Figure 6: Complex Plane for b'

Answer:

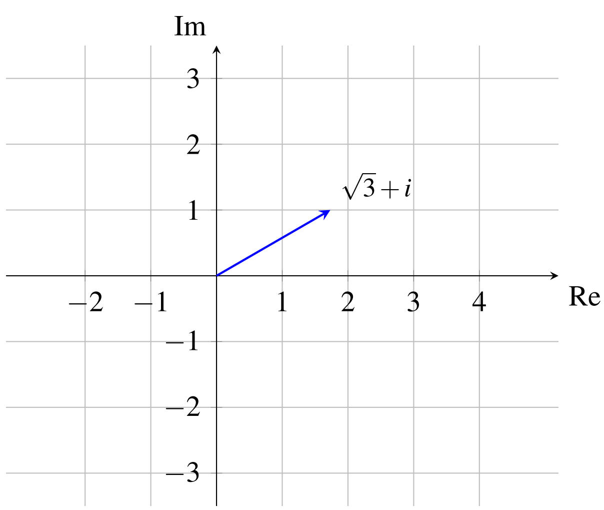

Figure 7: Complex Plane for a'

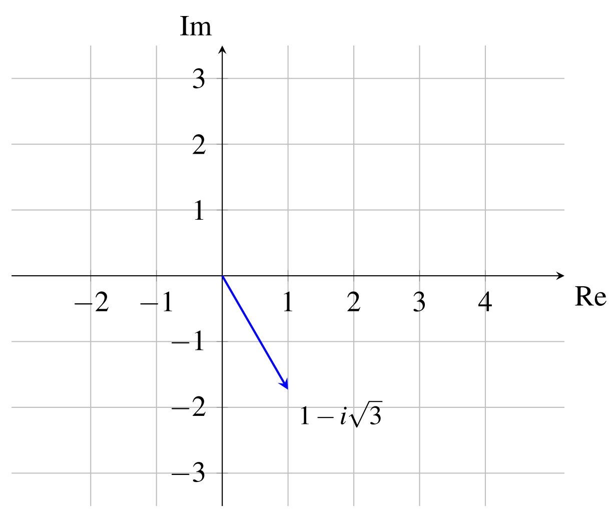

Figure 8: Complex Plane for b'

(c) For complex number $z = x + yi$ show that $|z| = \sqrt{z\bar{z}}$, where $\bar{z}$ is the complex conjugate of z.

Answer: We can follow the definition of complex conjugate and magnitude:

$$\sqrt{z\bar{z}} = \sqrt{(x+yi)(x-yi)} = \sqrt{x^2+y^2} = |z| \tag{1}$$

(d) Find ab, $a\bar{b}$, $a+\bar{a}$, $a-\bar{a}$, $\overline{ab}$, and $\bar{a}\bar{b}$.

Answer: We can evaluate these sequentially using the rules of complex algebra:

$$\begin{aligned}
ab &= (1-i\sqrt{3})(\sqrt{3}+i) = \sqrt{3}+\sqrt{3}+i(1-3) = 2\sqrt{3}-2i \\
a\overline{b} &= (1-i\sqrt{3})(\sqrt{3}-i) = \sqrt{3}-\sqrt{3}+i(-3-1) = -4i \\
a+\overline{a} &= 2 \\
a-\overline{a} &= -2i\sqrt{3} \\
\overline{ab} &= 2\sqrt{3}+2i \\
\overline{a}\overline{b} &= (1+i\sqrt{3})(\sqrt{3}-i) = \sqrt{3}+\sqrt{3}+i(3-1) = 2\sqrt{3}+2i
\end{aligned}$$

